Quantum Computing

Quantum States and Measurement

Lecture 1 Notes

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About these notes

These notes accompany a nine-lecture course on quantum computing designed and produced by the Lebanese Science Organization. Each chapter corresponds to one lecture. The lectures are concept-focused: they explain what a quantum state is, how entanglement, interference, and measurement combine to produce computation, and what is required to make that computation reliable. Linear algebra, complex numbers, and elementary probability are assumed.

Each chapter names its lecturer and date, opens with a short preface, develops its material with the derivations in place, and closes with a summary. Worked examples appear in numbered boxes, as do definitions, theorems, remarks, and common misconceptions. Works cited in the text are collected in a single reference list at the end of the document.

Conventions used throughout: qubits in a register are labelled from left to right, so that in $|ab\rangle$ the first qubit is a; the first qubit is the top wire in every circuit diagram; the computational basis of two qubits is ordered $|00\rangle, |01\rangle, |10\rangle, |11\rangle$; circuits are read from left to right, so a gate U_1 followed by a gate U_2 acts as the matrix product U_2U_1.

This chapter is exported individually from the full lecture notes (1 of 9). These notes are made available for open, public study. Reuse conditions will be stated separately by the Lebanese Science Organization.

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Chapter 1

Quantum States and Measurement

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This lecture introduces the two objects on which the rest of the course rests: the state of a qubit and the measurement that reads it. It begins with the experiment that forced both ideas into existence, the deflection of silver atoms by a magnet observed by Stern and Gerlach in 1922, and with the history of the word spin, which that experiment discovered without knowing it. A state is then a unit vector of complex amplitudes, and a measurement is a choice of basis that returns one basis element with a probability fixed by the corresponding amplitude. We examine what a state does and does not determine about the outcome, separate a superposition from classical ignorance and a global phase from a relative one, and meet the gates X, Z, and H, with which the first instance of interference appears.

1.1 The Stern-Gerlach experiment

Before any mathematics, we look at the experiment that made the mathematics necessary. Everything the rest of this lecture defines, the two-dimensional state, the choice of measurement basis, the rule that a measurement disturbs what it measures, was invented to describe the behaviour of a silver atom in a magnetic field. The experiment is also where the word spin comes from, and the history of that word is a useful warning about how easily a correct result acquires a wrong picture.

1.1.1 The experiment of 1922

In 1921, Otto Stern, then an assistant to Max Born in Frankfurt, proposed a test of a claim in the Bohr-Sommerfeld model of the atom [1]. In that model an electron orbits the nucleus, and an orbiting charge is a small magnet; the model further asserted that in an external magnetic field the orbit could take up only certain discrete orientations, a hypothesis called space quantization. Classical physics disagreed. A small magnet entering a field can point in any direction, so a beam of atoms sent through a field that is stronger on one side than the other should be deflected by a continuous range of amounts and arrive at a detector as a broadened smear. The Bohr model, as Stern read it, permitted exactly two orientations for the silver atom and so predicted two separated spots with nothing in between. Stern, who a few years earlier had vowed to quit physics if Bohr's model turned out to be right, expected the experiment to refute it.

Walther Gerlach, the experimentalist in the neighbouring institute, built the apparatus of Figure 1.1. Silver was evaporated in an oven at about 1000 °C; two slits shaped the escaping atoms

into a beam a few hundredths of a millimetre wide; the beam crossed a magnet whose pole pieces, a knife edge facing a groove, made the field far stronger near one pole than near the other; and the atoms condensed on a glass plate. The expected splitting was a fifth of a millimetre, and a misalignment of a hundredth of a millimetre spoiled a run. The deposit was so thin that it was at first invisible, and by Stern's own account it appeared only when he leaned over the plate: the sulphur in his cheap cigars turned the silver into black silver sulphide, and the trace developed like a photograph. In February 1922, after a year of failures and just after the two had agreed to give up, Gerlach obtained a clean result [2]. The beam split in two. There was no smear and no central spot. He telegraphed Stern, "Bohr is right after all," and sent Bohr a postcard with a photograph of the split beam and the message, "We congratulate you on the confirmation of your theory."

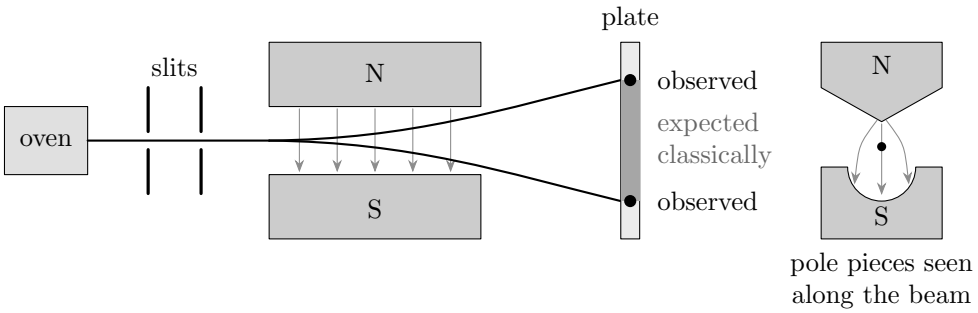


Figure 1.1: The Stern-Gerlach apparatus. Silver atoms leave an oven, are collimated into a narrow beam, and cross a magnet whose field is much stronger near the upper pole than near the lower one. Classical physics expects atoms with randomly oriented magnetic moments to be deflected by every amount up to a maximum, producing a continuous band on the plate. Two spots are observed. The inset shows the pole pieces as seen along the beam, a knife edge above and a groove below, which is what makes the field inhomogeneous; the dot is the beam.

The figures quoted above follow the historical account of Friedrich and Herschbach [3], which also records the cigar, the telegram, and the postcard. The oven ran at about $1000\,^{\circ}\mathrm{C}$; the two collimating slits were 0.03 millimetres wide; the magnet was 3.5 centimetres long with a field of about 0.1 tesla and a gradient of about 10 tesla per centimetre; the two traces were separated by about 0.2 millimetres; and the result was communicated to Bohr on 8 February 1922. Gerlach and Stern then measured the magnetic moment of the silver atom to within about ten percent of one Bohr magneton, $\mu_B = e\hbar/2m_e$ [4].

The congratulation was premature, in an instructive way. The Bohr-Sommerfeld reasoning that predicted two spots does not apply to silver: its single outer electron has no orbital angular momentum at all, and on the orbital picture the beam should not have split. What deflected the atoms was a property that nobody had yet named. In 1925 Wolfgang Pauli, to account for the structure of atomic spectra, attributed to the electron a "two-valuedness not describable classically" [5], and later that year George Uhlenbeck and Samuel Goudsmit proposed that the electron carries an intrinsic angular momentum and magnetic moment of its own, which they pictured as a rotation of the electron about its axis and called spin [6, 7]. Only in 1927 did Ronald Fraser point out that the Stern-Gerlach splitting of silver, hydrogen, and sodium had to be due to spin, since the orbital contribution in each case is zero [8]; the same year Phipps and Taylor repeated the experiment with hydrogen [9]. By a coincidence of the constants involved, explained in Remark 1.1, the magnitude of the electron's spin magnetic moment equals the one that Bohr's orbital model had predicted, so the number Gerlach and Stern measured agreed with a theory that did not apply [10]. The experiment was right; the theory it was believed to confirm was wrong; and the property it had discovered was spin, of which its discoverers had never heard. Stern was awarded the 1943 Nobel Prize in Physics for the molecular beam method that grew out of it.

Remark 1.1 The magnetic moment coincidence

The coincidence that let an inapplicable theory predict the right number is the following. A magnetic moment associated with an angular momentum has magnitude $g\,m\,\mu_B$, where m is the projection of the angular momentum along the field in units of $\hbar$ and g is a dimensionless factor. For orbital motion $g = 1$; the Bohr model assigned the silver atom $m = \pm 1$, excluding $m = 0$ on grounds that Bohr himself found uneasy, and so predicted moments $\pm\mu_B$. In fact the orbital angular momentum of the silver atom is zero, the relevant angular momentum is the electron's spin with $m = \pm\frac{1}{2}$, and the spin has $g \approx 2$, a value that emerged from Thomas's relativistic analysis in 1926 and is predicted by Dirac's equation of 1928. The product is again $\pm\mu_B$. The magnitude therefore agreed with Bohr's prediction, while the twofold splitting, which the Bohr model obtained only by discarding $m = 0$, is a genuine property of spin one half, for which $\pm\frac{1}{2}$ are the only values.

1.1.2 Sequential measurements

Strip the history away and what remains is a device with one input and two outputs. Atoms enter; each leaves through the upper port or the lower port, never anywhere else. Call the apparatus SG_z when its field points along the vertical axis z, and SG_x when the whole magnet is rotated by a right angle about the beam so that the field points along the horizontal axis x. Nothing about the two-way split depends on the orientation: SG_x also splits any beam into exactly two. Now connect several devices in series, blocking one output of each and feeding the other into the next, as in Figure 1.2. Four observations result.

(i) **Two outcomes:** Whatever the orientation of the apparatus and whatever the history of the beam, exactly two spots appear. There is never a continuum and never a third spot.

(ii) **Repeatability:** Atoms that left SG_z through the upper port and enter a second SG_z all leave it through the upper port. The first result is reproduced with certainty.

(iii) **A perpendicular measurement is random:** Atoms that left SG_z through the upper port and enter SG_x leave it half through one port and half through the other. If the second apparatus is rotated by an angle ϑ rather than a right angle, the fractions are $\cos^2(\vartheta/2)$ and $\sin^2(\vartheta/2)$.

(iv) **The previous result is erased:** Take the atoms from (iii) that left SG_x through its upper port, every one of which was earlier certified "up" by SG_z, and send them into a third SG_z. Half leave through the lower port. The intervening x measurement has destroyed the certainty that (ii) established.

The arrangement as drawn is an idealization that has served as the standard entry to quantum mechanics since Feynman's lectures [11, 12]; its predictions have been confirmed in many equivalent experiments with atoms, neutrons, and the polarization of light. Observations (i) to (iv) are the specification that the rest of this lecture must meet. Observation (i) says that the system has two distinguishable states for every orientation of the magnet. Observation (ii) says that a measurement leaves the system in the state corresponding to its outcome. Observation (iii) says that the states distinguished by one orientation are not the states distinguished by another, and it gives the numerical relation between them. Observation (iv) says that a measurement is not a passive reading of a value that was there all along. Sections 1.2 and 1.3 construct the smallest mathematical object with all four properties: a unit vector in $\mathbb{C}^2$, with one orthonormal basis for each orientation of the magnet.

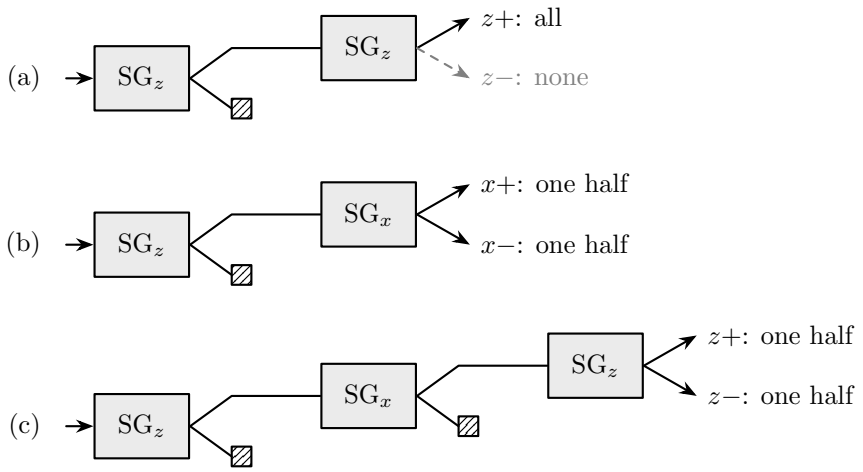


Figure 1.2: Sequential Stern-Gerlach experiments. A hatched square blocks one output. (a) A second apparatus in the same orientation confirms the first result with certainty. (b) An apparatus rotated by a right angle splits the certified beam in half. (c) A third apparatus in the original orientation finds the original result gone: half the atoms, every one of which left the first SG$_z$ through the upper port, now leave through the lower one.

Wrong Concept 1.1: Spin means the electron is spinning

Uhlenbeck and Goudsmit pictured the electron as a charged sphere rotating about an axis, and the name has stuck. The picture fails at once: for a sphere of the size then attributed to the electron to carry the observed angular momentum, its surface would have to move many times faster than light, an objection Lorentz raised as soon as he heard the proposal, and every later measurement has found the electron to be pointlike. Spin is not a rotation of anything. It is an intrinsic property of the electron that takes exactly two values along any chosen axis and that couples to magnetic fields and to rotations in the way an angular momentum does. In this course the word means no more than that: "spin up" and "spin down" name the two outcomes of a two-outcome measurement, and nothing turns.

Wrong Concept 1.2: The magnet sorts atoms that were already up or already down

Since the beam from the oven splits in half, it is tempting to say that half the atoms were up and half were down all along and that the magnet merely sorted them. For the beam straight from the oven this description is harmless, and **??** will make it precise. It fails at the second stage of Figure 1.2(b): every atom entering SG$_x$ has just been certified up, and the beam still splits in half. One might retreat to the claim that each atom carries a fixed answer for every orientation and that each apparatus reads off the relevant one. Then an atom certified up by SG$_z$ would still carry that answer after passing through SG$_x$, and the third apparatus in Figure 1.2(c) would return only up. It returns down half the time. A Stern-Gerlach apparatus therefore does something to the atom; it does not merely inspect it. The observation does not by itself exclude every account in which atoms carry hidden values, since one could imagine the x measurement disturbing a hidden z value. **??** returns to this question with pairs of atoms and shows that a large class of such accounts, those in which each atom carries its own answers undisturbed by anything done to a distant partner, is ruled out by experiment.

A silver atom is one physical system with this two-valued structure. The polarization of a

photon, the two lowest energy levels of a trapped ion or a superconducting circuit, and the spin of a single electron held in a semiconductor all have it too, and the last of these is used directly as a qubit in several laboratories, with a spin-dependent readout that is the modern descendant of Gerlach's magnet. The mathematics that follows is the same for all of them. We call any such system a qubit and now describe it.

1.2 States and measurement

Classical information is carried by a bit, which is either 0 or 1. Quantum information is carried by a qubit, whose state is a unit vector in a two-dimensional complex vector space. We write the two basis vectors as

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \tag{1.1}$$

and call them the computational basis. The notation $|\cdot\rangle$ is Dirac's ket, and it denotes a column vector. Nothing about it is new mathematics; it is a convenient label for a vector whose contents we care about physically.

Definition 1.1: Qubit state

A pure state of a qubit is a vector

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \qquad \alpha, \beta \in \mathbb{C}, \qquad |\alpha|^2 + |\beta|^2 = 1. \tag{1.2}$$

The complex numbers α and β are the amplitudes of $|\psi\rangle$ in the computational basis.

The normalization condition is not decoration. It is what makes the squared moduli $|\alpha|^2$ and $|\beta|^2$ a probability distribution over the two outcomes of a measurement, as the next definition states.

For each ket there is a bra, $\langle\psi| = |\psi\rangle^\dagger = \begin{pmatrix} \alpha^* & \beta^* \end{pmatrix}$, the conjugate transpose. The inner product of two states is

$$\langle\phi|\psi\rangle = \phi_0^*\psi_0 + \phi_1^*\psi_1, \tag{1.3}$$

and normalization reads $\langle\psi|\psi\rangle = 1$. Two states are orthogonal when their inner product is zero. The computational basis states are orthonormal: $\langle 0|0\rangle = \langle 1|1\rangle = 1$ and $\langle 0|1\rangle = 0$.

Definition 1.2: Measurement in the computational basis

Measuring the qubit $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$ in the computational basis yields the outcome 0 with probability $|\alpha|^2 = |\langle 0|\psi\rangle|^2$ and the outcome 1 with probability $|\beta|^2 = |\langle 1|\psi\rangle|^2$. After the outcome b is observed, the state of the qubit is $|b\rangle$.

This is the Born rule together with the projection rule. Three features deserve emphasis before we go further.

First, the outcome is random, and the state fixes only the distribution. Preparing many qubits identically in $|\psi\rangle$ and measuring each one produces a sequence of zeros and ones whose frequencies approach $|\alpha|^2$ and $|\beta|^2$.

Second, the state after the measurement is the basis vector corresponding to the outcome, not the original state. Measuring the same qubit a second time in the same basis returns the same result with certainty. Measurement disturbs every state that is not already one of the basis vectors.

Third, an amplitude is not a probability. The amplitude α may be negative or complex; only $|\alpha|^2$ is a probability. This distinction is the source of everything interesting in the course.

Observations (i) and (ii). Write $|0\rangle$ for the state of an atom that has left SG_z through the upper port and $|1\rangle$ for one that has left through the lower port. Then Definition 1.2 is SG_z itself. It has two outcomes and nothing in between, which is observation (i) of section 1.1, and an atom in the state $|0\rangle$ sent through a second SG_z gives the outcome 0 with probability $|\langle 0|0\rangle|^2 = 1$, which is observation (ii). Observations (iii) and (iv) involve a magnet in a different orientation, and describing them requires a different basis, which is the subject of the next section.

1.3 Superposition and phase

The state $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$ with both amplitudes nonzero is called a superposition of $|0\rangle$ and $|1\rangle$. The word does not mean that the qubit is secretly 0 or secretly 1 and we are ignorant of which. The distinction between a superposition and classical ignorance is exactly what this section makes precise.

1.3.1 Measurement bases

The computational basis is not privileged. Any orthonormal pair $\{|u\rangle, |v\rangle\}$ defines a measurement with two outcomes, with probabilities $|\langle u|\psi\rangle|^2$ and $|\langle v|\psi\rangle|^2$, and with the state afterwards being $|u\rangle$ or $|v\rangle$ accordingly. For an orthonormal basis $\{|u\rangle, |v\rangle\}$ the probabilities $|\langle u|\psi\rangle|^2 + |\langle v|\psi\rangle|^2 = 1$ because $|u\rangle\langle u| + |v\rangle\langle v| = \mathbb{I}$ and therefore $\langle\psi|\psi\rangle = \langle\psi|\,(|u\rangle\langle u| + |v\rangle\langle v|)\,|\psi\rangle = |\langle u|\psi\rangle|^2 + |\langle v|\psi\rangle|^2$. This is why orthonormality of the measurement basis is required.

The basis we will use most, after the computational basis, is

$$|+\rangle = \frac{1}{\sqrt{2}}\,(|0\rangle + |1\rangle)\,, \qquad |-\rangle = \frac{1}{\sqrt{2}}\,(|0\rangle - |1\rangle)\,. \tag{1.4}$$

These are orthonormal: $\langle +|-\rangle = \frac{1}{2}(1-1) = 0$. We call measurement in this basis the X measurement, for a reason that becomes clear in the next section. For a spin, $|+\rangle$ and $|-\rangle$ are the states of an atom that has left SG_x through its upper and lower ports, and measuring in this basis is what SG_x does.

Example 1.1 (Distinguishing $|+\rangle$ from $|-\rangle$)
Measured in the computational basis, both $|+\rangle$ and $|-\rangle$ give 0 and 1 with probability $\frac{1}{2}$ each. They are indistinguishable by that measurement. Measured in the $\{|+\rangle, |-\rangle\}$ basis, the state $|+\rangle$ gives the outcome $+$ with certainty and $|-\rangle$ gives the outcome $-$ with certainty. One change of basis turns two indistinguishable states into two perfectly distinguishable ones.

The point of the example is that the sign between the two amplitudes, which is invisible in the computational basis, is a real physical property, since some measurement reveals it with certainty.

The same two lines account for the remaining Stern-Gerlach observations. An atom in the state $|0\rangle$ enters SG_x; the probabilities of the two ports are $|\langle +|0\rangle|^2 = |\langle -|0\rangle|^2 = \frac{1}{2}$, which is observation (iii), and the atom leaves in the state $|+\rangle$ or $|-\rangle$, not in the state $|0\rangle$. If it then enters a third SG_z, the probability of the lower port is $|\langle 1|+\rangle|^2 = \frac{1}{2}$, which is observation (iv). The certainty established by the first apparatus was not hidden inside the atom and then overwritten; it was a property of the state $|0\rangle$, and the atom is no longer in that state.

Normalization and probabilities. Let $|\psi\rangle = \frac{1}{\sqrt{3}}\,|0\rangle + \sqrt{\frac{2}{3}}\,\mathrm{e}^{\mathrm{i}\pi/4}\,|1\rangle$. Then $|\alpha|^2 = \frac{1}{3}$ and $|\beta|^2 = \frac{2}{3}$, so the state is normalized and measurement in the computational basis gives 0 with probability $\frac{1}{3}$

and 1 with probability $\frac{2}{3}$. The phase $e^{i\pi/4}$ plays no role in these probabilities. It does affect the probabilities in the $\{|+\rangle, |-\rangle\}$ basis:

$$\langle +|\psi\rangle = \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{3}} + \sqrt{\frac{2}{3}}\, e^{i\pi/4}\right), \qquad |\langle +|\psi\rangle|^2 = \frac{1}{2}\left(\frac{1}{3} + \frac{2}{3} + 2\cdot\frac{\sqrt{2}}{3}\cos\frac{\pi}{4}\right) = \frac{1}{2} + \frac{1}{3} = \frac{5}{6}. \quad (1.5)$$

Since $|\langle +|\psi\rangle|^2 = \frac{1}{2}|\alpha+\beta|^2 = \frac{1}{2}\left(|\alpha|^2 + |\beta|^2\right) + \mathrm{Re}(\alpha^*\beta)$, the cross term $\mathrm{Re}(\alpha^*\beta)$ is where the relative phase enters. Changing $\pi/4$ to $5\pi/4$ changes the probability of $+$ to $\frac{1}{6}$.

1.3.2 Global and relative phase

Multiply a state by a complex number of modulus one: $|\psi\rangle \mapsto e^{i\gamma}|\psi\rangle$. Every measurement probability is unchanged, since $|e^{i\gamma}\langle u|\psi\rangle|^2 = |\langle u|\psi\rangle|^2$ for every $|u\rangle$. The factor $e^{i\gamma}$ is called a global phase, and two states differing by a global phase are physically the same state.

Explicitly, for any basis vector $|u\rangle$, $|\langle u|e^{i\gamma}\psi\rangle|^2 = |e^{i\gamma}|^2|\langle u|\psi\rangle|^2 = |\langle u|\psi\rangle|^2$. Since this holds for every measurement, no experiment distinguishes $|\psi\rangle$ from $e^{i\gamma}|\psi\rangle$. Unitary evolution, introduced in section 1.4, also respects this: $U(e^{i\gamma}|\psi\rangle) = e^{i\gamma}U|\psi\rangle$ by linearity, so the global phase is carried along and remains unobservable at every later time.

Now write a general normalized state as

$$|\psi\rangle = \cos\tfrac{\theta}{2}\,|0\rangle + e^{i\varphi}\sin\tfrac{\theta}{2}\,|1\rangle, \qquad 0 \leqslant \theta \leqslant \pi, \quad 0 \leqslant \varphi < 2\pi, \quad (1.6)$$

which is always possible after discarding a global phase. The angle φ is the relative phase between the two amplitudes. Unlike a global phase, a relative phase is physical: $|+\rangle$ and $|-\rangle$ have $\theta = \pi/2$ and differ only in $\varphi = 0$ versus $\varphi = \pi$, and Example 1.1 shows that a measurement distinguishes them.

Wrong Concept 1.3: Superposition is ignorance about a hidden bit

Suppose someone flips a fair coin and prepares $|0\rangle$ or $|1\rangle$ according to the result, without telling us which. Measured in the computational basis, this preparation gives 0 and 1 with probability $\frac{1}{2}$ each, exactly like $|+\rangle$. But measured in the $\{|+\rangle, |-\rangle\}$ basis it gives $+$ and $-$ with probability $\frac{1}{2}$ each, whereas $|+\rangle$ gives $+$ with certainty. The coin-flip preparation and the superposition $|+\rangle$ agree in one basis and disagree in another, so they are different physical situations. A superposition is not a hidden classical bit.

The coin-flip preparation yields $|0\rangle$ with probability $\frac{1}{2}$ and $|1\rangle$ with probability $\frac{1}{2}$. Measured in the $\{|+\rangle, |-\rangle\}$ basis, $|0\rangle$ gives $+$ with probability $|\langle +|0\rangle|^2 = \frac{1}{2}$, and so does $|1\rangle$. Averaging over the coin, the outcome $+$ has probability $\frac{1}{2}$. For the state $|+\rangle$ the outcome $+$ has probability 1. This completes the argument in the box on superposition and ignorance above. In **??** the coin-flip preparation will be described by a density matrix, and the distinction made here will be read off from its off-diagonal entries.

1.3.3 The Bloch sphere

The parametrization (1.6) identifies every pure qubit state, up to global phase, with a point on the unit sphere in three dimensions with polar angle θ and azimuthal angle φ. This is the Bloch sphere. The north pole is $|0\rangle$, the south pole is $|1\rangle$, and $|+\rangle$ and $|-\rangle$ sit on the equator at $\varphi = 0$ and $\varphi = \pi$. Orthogonal states are antipodal points. A measurement in a given basis asks which of two antipodal points the state is closer to, in the sense that the probability of the outcome $|u\rangle$ is $\cos^2(\vartheta/2)$ where ϑ is the angle on the sphere between the state and $|u\rangle$. For a spin the sphere is not

a metaphor: the point (θ, φ) is the direction in space along which a Stern-Gerlach magnet would send the atom through its upper port with certainty, and the $\cos^2(\vartheta/2)$ rule is observation (iii) of section 1.1 with ϑ the angle by which the magnet was rotated.

The $\cos^2(\vartheta/2)$ law in observation (iii) is the Born rule of section 1.3.1 applied to a basis rotated by ϑ: with the state $|0\rangle$ and the basis vector $|u\rangle = \cos\frac{\vartheta}{2}|0\rangle + \sin\frac{\vartheta}{2}|1\rangle$ of (1.6), one has $|\langle u|0\rangle|^2 = \cos^2(\vartheta/2)$, which is one half at $\vartheta = \pi/2$. The halving of the angle is the signature of spin one half. Rotating the magnet by a full turn returns the same outcome probabilities, but the parametrization (1.6) shows that the state vector itself acquires a factor -1, which section 1.3.2 identifies as a global phase and hence as unobservable on a single qubit.

We use the Bloch sphere for geometric intuition only. We will not convert between Bloch coordinates and amplitudes in these notes.

1.4 Unitary gates

Between measurements, an isolated qubit evolves linearly, and the evolution must preserve normalization so that probabilities continue to sum to one. A linear map on $\mathbb{C}^2$ that preserves the norm of every vector is a unitary matrix: a 2×2 matrix U with $U^\dagger U = \mathbb{I}$. Every such matrix is reversible, since $U^{-1} = U^\dagger$. In the circuit model of computation, these unitaries are called gates.

If $U^\dagger U = \mathbb{I}$ then $\langle U\psi|U\psi\rangle = \langle\psi|\,U^\dagger U\,|\psi\rangle = \langle\psi|\psi\rangle$. Conversely, a linear map on $\mathbb{C}^2$ that preserves every norm preserves every inner product, by the polarization identity, and therefore satisfies $U^\dagger U = \mathbb{I}$. This is why the gates of a quantum computer are exactly the unitary matrices.

Three gates recur throughout the course.

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \tag{1.7}$$

Their actions on the basis states are all one needs to remember:

$$X|0\rangle = |1\rangle, \qquad X|1\rangle = |0\rangle, \tag{1.8}$$
$$Z|0\rangle = |0\rangle, \qquad Z|1\rangle = -|1\rangle, \tag{1.9}$$
$$H|0\rangle = |+\rangle, \qquad H|1\rangle = |-\rangle. \tag{1.10}$$

The gate X is the quantum version of NOT: it exchanges $|0\rangle$ and $|1\rangle$. On the other basis it acts by a sign alone, $X|+\rangle = |+\rangle$ and $X|-\rangle = -|-\rangle$, so $|+\rangle$ and $|-\rangle$ are the eigenvectors of X with eigenvalues $+1$ and -1; this is why measurement in the $\{|+\rangle, |-\rangle\}$ basis is called the X measurement, just as measurement in the computational basis, whose vectors are the eigenvectors of Z, is called the Z measurement. The gate Z leaves $|0\rangle$ alone and reverses the sign of $|1\rangle$; on a superposition it changes the relative phase by π, so $Z|+\rangle = |-\rangle$ and $Z|-\rangle = |+\rangle$. The gate H, the Hadamard gate, exchanges the computational basis with the $\{|+\rangle, |-\rangle\}$ basis, and it is its own inverse: $H^2 = \mathbb{I}$.

The Hadamard gate is the operation that makes relative phase observable. Measuring in the $\{|+\rangle, |-\rangle\}$ basis has the same outcome statistics as applying H and then measuring in the computational basis, because $H|+\rangle = |0\rangle$ and $H|-\rangle = |1\rangle$; the two procedures differ only in the state they leave behind, $|+\rangle$ or $|-\rangle$ in the first case and $|0\rangle$ or $|1\rangle$ in the second. Whenever a circuit ends with H followed by a computational-basis measurement, it is performing an X measurement on the state that entered the H, and the statistics of an X measurement are sensitive to the relative phase of that state.

Example 1.2 (Interference through H, Z, H)

Start in $|0\rangle$. Apply H to obtain $|+\rangle$. Apply H again and the state returns to $|0\rangle$: measurement gives 0 with certainty. Now start in $|0\rangle$, apply H to obtain $|+\rangle$, apply Z to obtain $|-\rangle$, and apply H to obtain $|1\rangle$: measurement gives 1 with certainty. The only difference between the two circuits is a relative sign introduced in the middle, and that sign flips the final outcome deterministically. The two amplitudes of the intermediate state recombine constructively in one case and destructively in the other. This is interference, and it is the mechanism every quantum algorithm in this course relies on.

In matrix language the two examples are the identities $HH = \mathbb{I}$ and $HZH = X$. Direct multiplication gives

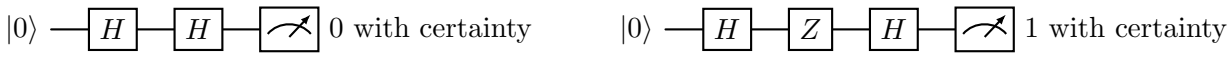
$$HH = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \mathbb{I}, \tag{1.11}$$

and

$$HZH = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix} = X. \tag{1.12}$$

The second identity is worth remembering: conjugating Z by H produces X, which is the algebraic statement that H exchanges the roles of bit value and phase. Similarly $HXH = Z$, and X, Z, H are each unitary and each their own inverse. The remaining Pauli matrix is $Y = \mathrm{i}XZ = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}$; it appears in later chapters as the third error type but is not needed for the arguments of this one.

Circuit notation. The two circuits of Example 1.2 are drawn as follows. Wires carry qubits from left to right, boxes are gates, and the meter is a computational-basis measurement.



Because circuits are read left to right while matrices act on the ket from the left, the gate applied first is written rightmost in the product. The second circuit therefore computes $HZH\,|0\rangle = X\,|0\rangle = |1\rangle$; the order happens not to matter here because HZH reads the same in both directions, but it matters for every sequence that does not.

Remark 1.2 Evolution versus measurement

Unitary evolution is deterministic, reversible, and linear. Measurement is random, irreversible, and produces a classical outcome. Every quantum computation consists of unitary evolution followed by measurement, and the art of designing one is to arrange the unitary part so that the final measurement is informative. We saw the smallest instance of this in Example 1.2: two amplitudes that are individually uninformative when measured directly are made to interfere so that the measurement becomes certain.

1.5 Summary

A Stern-Gerlach magnet in any orientation has exactly two outcomes, repeats its own result, and erases the result of a magnet in a perpendicular orientation; spin is the name of this two-valued property, and it is not a rotation. A qubit state is a unit vector of amplitudes. Measurement in a basis yields one basis element with probability given by the squared modulus of the corresponding amplitude, and leaves the qubit in that basis element. Which basis we measure in matters. A global phase is unobservable; a relative phase is observable by choosing the right basis. A superposition is not classical ignorance about a hidden bit, because some measurement distinguishes the two. Gates are unitary matrices, and the Hadamard gate is the one that turns relative phase into a measurable bit. Interference, in which amplitudes add or cancel, is what makes the final measurement informative.

Homework 1, part A, can be attempted after this lecture.

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